Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
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Term-Work

Of

Web Development (PCS-693)

Submitted in partial fulfilment of the requirement for the VI semester

Bachelor of Technology

(Computer Science & Engineering)

By

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2361453

Under the Guidance of

Ms. Megha Upreti

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

GRAPHIC ERA HILL UNIVERSITY, BHIMTAL CAMPUS

SATTAL ROAD, P.O. BHOWALI

2025-2026

CERTIFICATE

The Term work of Web Development, being submitted by Rishita joshi D/O Mr. Sunil Kumar Joshi, Enrollment no PV-23610453, Roll no. 2361453 to **Graphic Era Hill University Bhimtal Campus** for the award of bona fide work carried out by her. She has worked under our guidance and supervision and fulfilled the

(……) **(……)**

Ms. Megha Upreti **HOD, CSE Dept.**

ACKNOWLEDGEMENT

I take immense pleasure in thanking Ms. Megha Upreti **(Assistant Professor, CSE, GEHU Bhimtal Campus f**or allowing us to carry out this practical work under their excellent and optimistic supervision. This has all been possible due to their novel inspiration, able guidance and useful suggestions that have helped me in developing my subject concepts as a student.

Words are inadequate in offering my thanks to GOD for providing me everything that I need. I again want to extend thanks to our President **Prof. (Dr.) Kamal Ghanshala** for providing us all infrastructure and facilities to work in need without which this work could not be possible.

Many thanks to **Prof. (Col.) Anil Kumar Nair (Retd.),** (Director Gehu Bhimtal), other faculties for their insightful comments, constructive suggestions, valuable advice, and time during the review process.

STUDENT'S DECLARATION

I Rishita Joshi hereby declare the work, which is being presented in the term- work, entitled Web Development **Lab** in partial fulfillment of the requirement for the award of the degree **B.Tech** in the session **2025-2026**, is an authentic record of my own work carried out under the supervision of Ms. Megha Upreti **.**

The matter embodied in this term-work has not been submitted by me for the award of any other degree.

Date:

…….

(Rishita Joshi)

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S.NO	DATE	TITLE	PAGE NO.	REMARK
1.		Write a HTML code to draw the following List.		
2.		Create a web page to print the following table.		
3.		Using table related tags align the image with hyperlinks		
4.		Create a web page having frames as follows: The frame which includes Objective, Personal Information ect. are the hyperlinks. Display the relevant information in the next frame on selecting the link. The colour scheme of hyperlinks should be as follows: Default-green Active-red Visited - blue The information should be well formatted. Follow the instructions mentioned in the above format.		
5.		Demonstrate the use of <IFRAME> tag using suitable example		
6.		Create a program to run YouTube video in a web page.		
7.		Create a web page of customer profile for data entry of customers in a Hotel, The profile should include : Name, Address, Age, gender, Room Type (A/C, Non-A/C or Deluxe), Type of payment (Cash, Credit/Debit Card or Coupons).		
8.		Demonstrate the use of following HTML5. I. <Video> II. <Audio> III. <Header> IV. <Footer> V. <Nav> VI. <Embed> VII. <Datalist> VIII. <Bdi> IX. <Article> X. <Output>		

9.		Create Box Shadow and text Shadow using CSS3. 2. Create Rounded Corners using css3. 3. Create a web page to show fixed background image (this image will not scroll with the rest of the page) 4. Create a web page to position a background image and repeat the image horizontally or vertically. 5. Create a web page to resize the background of web page. 6. Create a web page to show all hyperlinks with following specification: Default color is pink. Active color is blue Visited color is Green Hyperlink should be without underline. 7. Design a web page to show border image. 8. Design a web page to show different border style		
10.		Design a web page using CSS which includes the following: • Use different font styles • Set background image for both the page and single elements on page. • Control the repetition of image with background-repeat property • Add customized cursors for links. • Work with layers.		
11.		Create a web page to show transition effect in such a way so that elements gradually change from one style to another style.		
12.		Design a web page to validate credit card number according to the below specification. Following tables outlines the major credit cards you might want to validate, along with their allowed prefixes and lengths.		
13.		Write a java script to validate the following fields in a registration page: 1. Name (should contains alphabets and the length should not be less than 6 characters) 2. Password(should not be less than 6 characters) 3. E-mail(should not contain invalid addresses)		

14.		Write a java script program to "Wish a user" at different hours of a day.		
15.		Design a web page that self modifying itself after every one minute.		
16.		Write an code for web application, which accepts the birth date from the user in a textbox and display the day of the week in a message box on the click of a button.		
17.		Write a script that inputs a telephone number as a string in the form (555)555-555. The script should use strings method split to extract the area code as token and the last four digits of the phone numbers as a token. Display the area code in one test field and the seven digit phone number in another text field.		
18.		Create a web page that applies the invert filter to an image if the user moves the mouse over it.		
19.		Write an HTML Page to scroll the text message from right to left at the status bar of browser window.		
20.		Design a web page to perform survey on four different model of Maruti (Maruti - K10, Zen-Astelo, Wagnor, Maruti- SX4) owned by person living in four metro cities(Delhi, Mumbai, Chennai &Kolkatta). Display tabulated report like format given below		
21.		Create a Login form having User-Id and Password fields. After submitting the form match the user-id and password with existing user-id and password. If user-id and password match a new welcome window should be appear.		

PROGRAM OBJECTIVE:

1. **Write a HTML code to draw the following List:**

Lists in HTML

1. Apple
2. Banana
3. Cherry

- Mango
- Pineapple
- Watermelon

CODE -

```
<!doctype html>
<html>
 <head>
     <title>List</title>

 </head>
 <body>
     <h1>Lists in HTML</h1>
     <ol>
         <li> apple </li>
         <li>Banana</li>
         <li>Cherry</li>
     </ol>
     <ul>
         <li>mango</li>
         <li>Pineapple
         </li>
         <li>Watermelon</li>
     </ul>
 </body>
 </html>
```

OUTPUT:-

Lists in HTML

1. apple
2. Banana
3. Cherry

- mango
- Pineapple
- Watermelon

PROGRAM OBJECTIVE:

1. Create a web page to print the following table.

List of Course-wise Subjects

Sr. No.	Course	Subject	Marks		Category		Practical / Theory
			Internal	External	Internal	External	
1	MBA	Management Accounting	30	70	-	✓	Theory
		Information Technology	30	70	-	✓	T and P
		Basics of Marketing	30	70	-	✓	Theory
		E-Commerce	50	-	✓	-	Theory
2	MCM	Visual Basic	30	70	-	✓	T and P
		Internet Technology	30	70	-	✓	T and P
		Network Technology	30	70	-	✓	Theory
		VB.Net	30	70	-	✓	T and P
		Linux	30	70	-	✓	T and P
		ISA	50	-	✓	-	Theory

CODE:

```
<!Doctype html>
<body>
<h2 style="color : blue; text-align:center;">List of Course-Wise Subjects</h2>
  <table border="n" style= "width:100%; text-align: center; background-color: rgb(170, 210, 239);">
    <tr>
      <th rowspan="2">Sr. <br>No.</th>
      <th rowspan="2">Course</th>
      <th rowspan="2">Subject</th>
      <th colspan="2">Marks</th>
      <th colspan="2">Category</th>
      <th rowspan="2">Practical/ <br> Theory</th></tr>
<tr>
  <th>Internal</th>
  <th>External</th>
  <th>Internal</th>
  <th>External</th></tr>
<tr>
  <td rowspan="4">1</td>
  <td rowspan="4"><br>MBA</td>
  <td>MAnagement Accounting</td>
  <td>30</td>
  <td>70</td>
  <td>-</td>
  <td>✓</td>
  <td>Theory</td></tr>
<tr>
  <td>Information Technology</td>
  <td>30</td>
  <td>70</td>
  <td>-</td>
  <td>✓</td>
  <td>T and P</td></tr>
<tr>
  <td>Basics of Marketing</td>
  <td>30</td>
  <td>70</td>
  <td>-</td>
  <td>✓</td>
  <td>Theory</td></tr>
<tr> <td>E-Commerce</td>
  <td>50</td>
```

```
    <td>✓</td>
    <td>-</td>
    <td>Theory</td></tr>
<tr>
    <td rowspan="6">2</td>
    <td rowspan="6">MCM</td>
    <td>Visual Basic</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>T and P</td></tr>
<tr>
    <td>Internet Technology</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>T and P</td></tr>
<tr>
    <td>Network Technology</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>Theory</td></tr>
<tr>
    <td>VB. Net</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>Theory</td>     </tr>
<tr>
    <td>Linux</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>Theory</td> </tr>
<tr>
    <td>Software Testing</td>
    <td>30</td>
    <td>70</td>
    <td>-</td>
    <td>✓</td>
    <td>Theory</td> </tr>
</table>
</body>
```

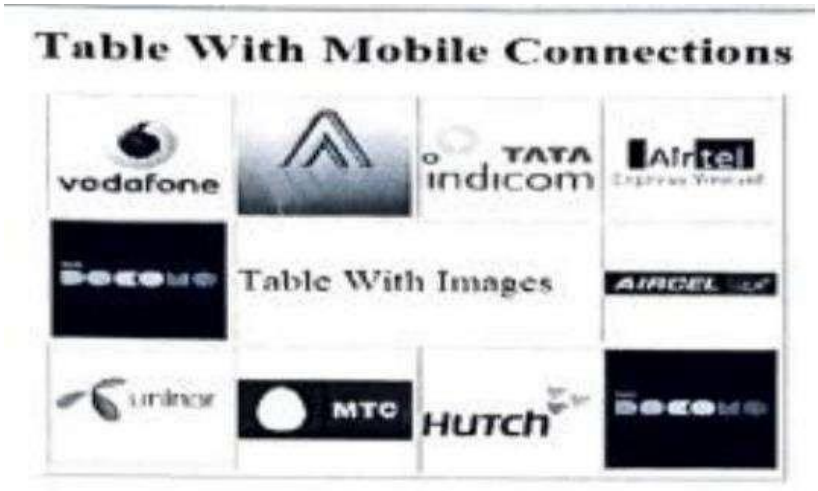
OUTPUT:

List of Course-Wise Subjects

Sr. No.	Course	Subject	Marks		Category		Practical/ Theory
			Internal	External	Internal	External	
1	MBA	MAnagement Accounting	30	70	-	✓	Theory
		Information Technology	30	70	-	✓	T and P
		Basics of Marketing	30	70	-	✓	Theory
		E-Commerce	50	-	✓	-	Theory
2	MCM	Visual Basic	30	70	-	✓	T and P
		Internet Technology	30	70	-	✓	T and P
		Network Technology	30	70	-	✓	Theory
		VB. Net	30	70	-	✓	Theory
		Linux	30	70	-	✓	Theory
		Software Testing	30	70	-	✓	Theory

PROGRAM OBJECTIVE:

2. Using table related tags align the image with hyperlinks



CODE:

```
    <!DOCTYPE html>
<html>
<head>
   <title>Mobile Connections</title>
   <style>
      table {
         margin: auto;
         border-collapse: collapse;
      }
      td {
         border: 1px solid black;
         padding: 10px;
         text-align: center;
      }
      img {
         width: 120px;
         height: 80px;
      }
   </style>
</head>

<body>
<h2 style="text-align: center;">Table With Mobile Connections</h2>
<table>
   <tr>
      <td>
         <a href="https://www.vodafone.in" target="_blank"><img src="../images/voda.jpeg" alt="Vodafone"> </a>
      </td>
      <td>
         <a href="https://www.airtel.in" target="_blank"><img src="../images/airtel.png" alt="Airtel"></a>
      </td>
      <td>
         <a href="https://www.tatadocomo.com" target="_blank"> <img src="../images/tata.png" alt="Tata"> </a>
      </td>
      <td>
```

```
            <a href="https://www.jio.com" target="_blank"> <img src="../images/jio.png" alt="Jio"> </a>
        </td>
    </tr>

    <tr>
        <td>
            <a href="https://www.bsnl.co.in" target="_blank"> <img src="../images/bsnl.png" alt="BSNL"> </a>
        </td>
        <td colspan="2">
            <b>Table With Images</b>
        </td>
        <td>
            <a href="https://www.aircel.com" target="_blank">  <img src="../images/aircel.png" alt="Aircel">
            </a>
        </td>
    </tr>

    <tr>
        <td>
            <a href="https://www.uninor.in" target="_blank"> <img src="../images/uninor.jpeg " alt="Uninor">
            </a>
        </td>
        <td>
            <a href="#"><img src="../images/mtnl.png" alt="MTNL">
            </a>
        </td>
        <td>
            <a href="#"> <img src="../images/hutch.png" alt="Hutch"> </a>
        </td>
        <td>
            <a href="#"> <img src="../images/bsnl.png" alt="BSNL"> </a>
        </td>
    </tr>
</table>
</body>
</html>
```

OUTPUT :

PROGRAM OBJECTIVE:

4. Create a web page having frames as follows:

Your Name Punch line etc.		
Objective Personal Information Family Information Educational Information Experience Achievements Other	Display information here of selected link	Display relevant images here
Your contact details		

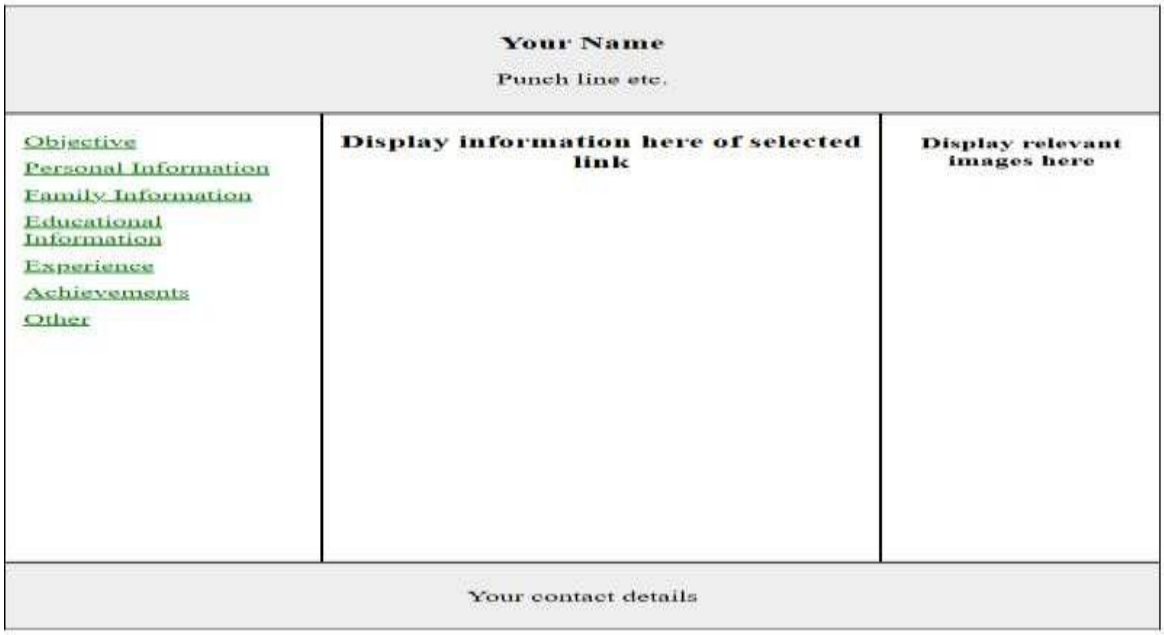
The frame which includes Objective, Personal Information ect. are the hyperlinks. Display the relevant information in the next frame on selecting the link. The colour scheme of hyperlinks should be as follows:
Default-green , Active-red ,Visited - blue

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Frame Layout (Single File)</title>
   <style>

      .header, .footer {
         text-align: center;
         padding: 10px;
         border: 1px solid black;
         background: #eee;
      }
      .container {
         display: flex;
         height: 70vh;
      }
      .menu {
         width: 25%;
         border: 1px solid black;
         padding: 10px;
      }
      .content {
         width: 50%;
         border: 1px solid black;
      }
      .image {
         width: 25%;
         border: 1px solid black;
      }
      iframe {
         width: 100%;
         height: 100%;
         border: none;
      }
      a:link { color: green; }
      a:active { color: red; }
      a:visited { color: blue; }
```

```
        a {
            display: block;
            margin: 10px 0;
        }
    </style>
</head>
<body>
<!-- Header -->
<div class="header">
    <h3>Your Name</h3>
    <p>Punch line etc.</p>
</div>

<!-- Middle Section -->
<div class="container">

    <!-- Left Menu -->
    <div class="menu">
        <a href="objective.html" target="contentFrame">Objective</a>
        <a href="personal.html" target="contentFrame">Personal Information</a>
        <a href="family.html" target="contentFrame">Family Information</a>
        <a href="education.html" target="contentFrame">Educational Information</a>
        <a href="experience.html" target="contentFrame">Experience</a>
        <a href="achievement.html" target="contentFrame">Achievements</a>
        <a href="other.html" target="contentFrame">Other</a>
    </div>
    <!-- Center Content -->
    <div class="content">
        <iframe name="contentFrame" srcdoc="<h3 style='text-align:center'>Display information here of selected
link</h3>"></iframe>
    </div>

    <!-- Right Image -->
    <div class="image">
        <iframe srcdoc="<h4 style='text-align:center'>Display relevant images here</h4>"></iframe>
    </div>
</div>

<div class="footer">
    <p>Your contact details</p>
</div>
</body>
</html>
```

OUTPUT:



PROGRAM OBJECTIVE:

5. Demonstrate the use of <IFRAME> tag using suitable example

CODE:

```
 <!DOCTYPE html>
<html>
<head>
   <title>Iframe Example</title>
</head>

<body>

<h2>Demonstration of IFRAME</h2>

<!-- iframe to display another webpage -->
<iframe src="https://www.wikipedia.org" width="600" height="300"></iframe>
<br><br>

</body>
</html>
```

OUTPUT:

PROGRAM OBJECTIVE:

6. Create a program to run YouTube video in a web page.

CODE:

```
<!DOCTYPE html>
<html >
<head>
   <title>YouTube Embed</title>
<style>
   body {
      text-align: center;
      background-color: rgb(235, 195, 179);
   }
   </style>
</head>
<body>
   <h3>YouTube Video Embed Example</h3>
   <iframe width="500" height="315"
   src="https://www.youtube.com/embed/CtQPknZzttw?si=XOSb73Uobb8mmYC3"
 title="YouTube video player" frameborder="0" allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture; web-share"
    referrerpolicy="strict-origin-when-cross-origin" allowfullscreen></iframe>
</body>
</html>
```

OUTPUT:

PROGRAM OBJECTIVE:

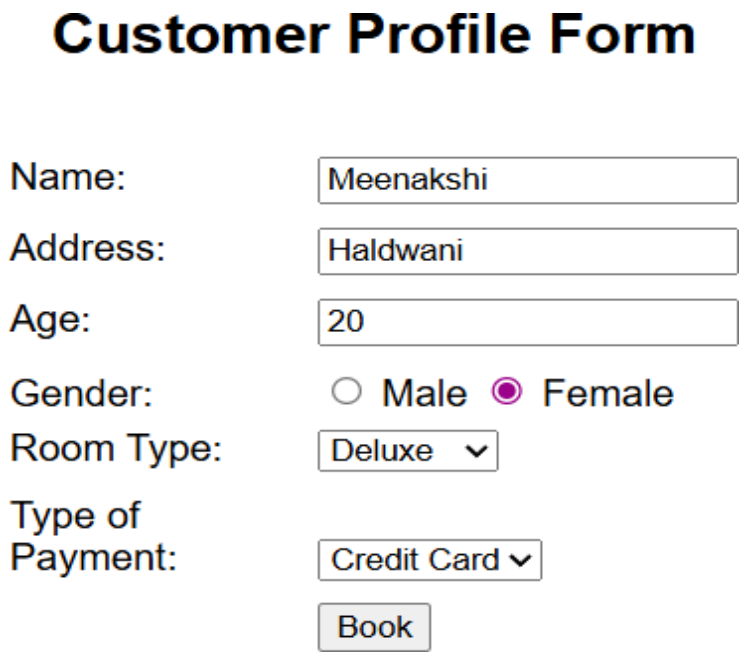
7. Create a web page of customer profile for data entry of customers in a Hotel, The profile should include :
Name, Address, Age, gender, Room Type (A/C, Non-A/C or Deluxe), Type of payment (Cash, Credit/Debit Card or Coupons).

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Hotel Customer Profile</title>
   <style>
      body {
         font-family: Arial;
         text-align: center;
      }
      form {
         display: inline-block;
         text-align: left;
         padding: 20px;
      }
      label {
         display: inline-block;
         width: 120px;
      }
      input, select {
         margin-bottom: 10px;
      }
      button {
         margin-top: 10px;
      }
   </style>
</head>

<body>
<h2>Customer Profile Form</h2>
<form>
   <!-- Name -->
   <label>Name:</label>
   <input type="text" name="name" required><br>

   <!-- Address -->
   <label>Address:</label>
   <input type="text" name="address"><br>

   <!-- Age -->
   <label>Age:</label>
   <input type="number" name="age"><br>

   <!-- Gender -->
   <label>Gender:</label>
   <input type="radio" name="gender"> Male
   <input type="radio" name="gender"> Female

  <br>
```

```
    <!-- Room Type -->
    <label>Room Type:</label>
    <select name="room">
        <option>A/C</option>
        <option>Non-A/C</option>
        <option>Deluxe</option>
    </select><br>

    <!-- Payment Type -->
    <label>Type of Payment:</label>
    <select name="payment">
        <option>Cash</option>
        <option>Credit Card</option>
        <option>Debit Card</option>
        <option>Coupons</option>
    </select><br>

    <!-- Button -->
    <center><input type="submit" value="Book"   </center>

</form>
</body>
</html>
```

OUTPUT:



PROGRAM OBJECTIVE:

8. Demonstrate the use of following HTML5.

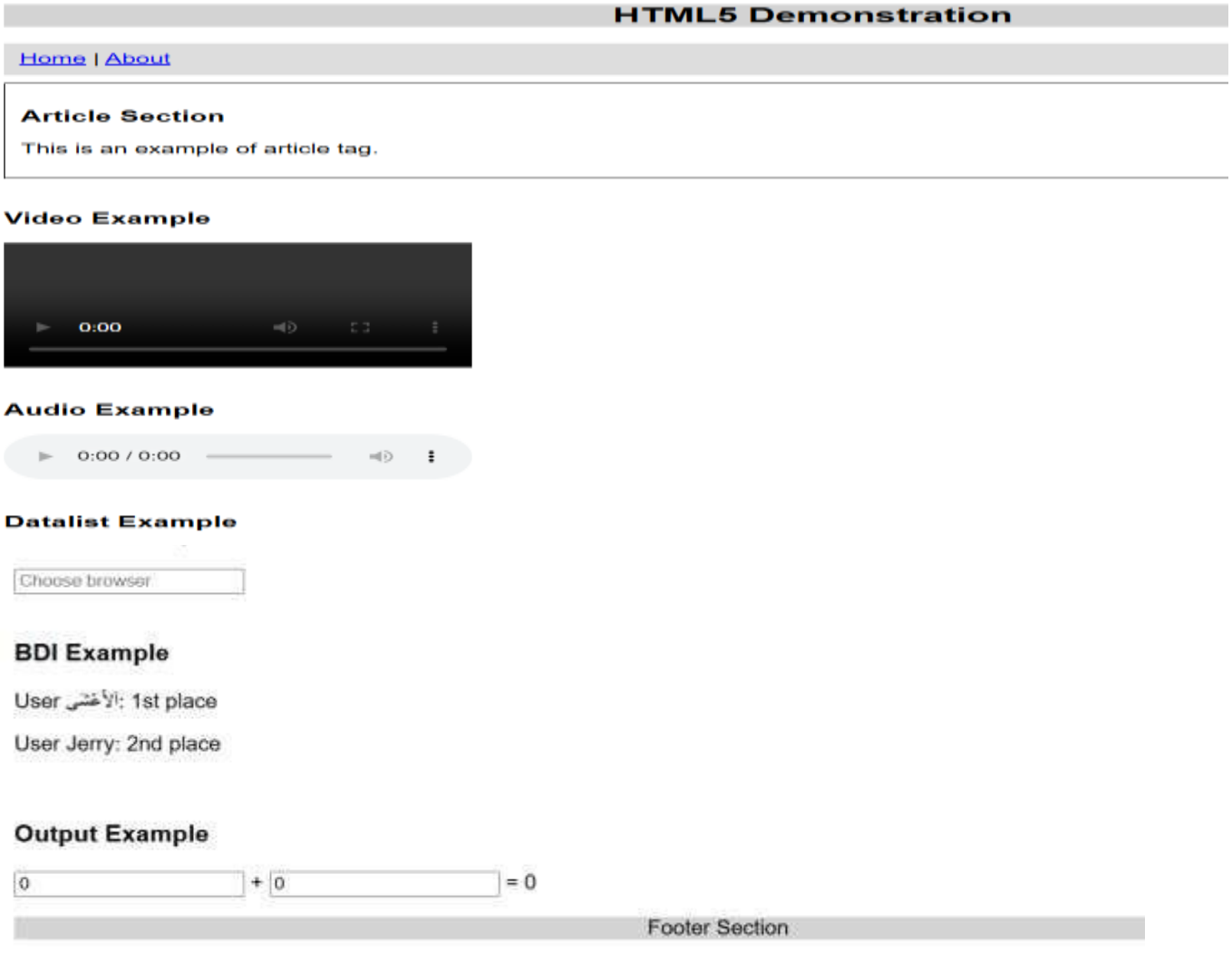
I <Video>
II <Audio>
III <Header>
IV <Footer>
V <Nav>
VI <Embed>
VII <Datalist>
VIII <Bdi>
IX <Article>
X <Output>

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>HTML5 Tags Demo</title>
   <style>
      body { font-family: Arial; margin: 20px; }
      header, footer { background: lightgray; text-align: center;}
      nav { background: #ddd; padding: 10px; }
      article { border: 1px solid black; padding: 10px; margin-top: 10px; }
   </style>
</head>
<body>
<!-- HEADER -->
<header>
   <h2>HTML5 Demonstration</h2>
</header>
<!-- NAV -->
<nav>
   <a href="#">Home</a> |
   <a href="#">About</a>
</nav>

<!-- ARTICLE -->
<article>
   <h3>Article Section</h3>
   <p>This is an example of article tag.</p>
</article>

<!-- VIDEO -->
<h3>Video Example</h3>
<video width="300" controls>
   <source src="video.mp4" type="video/mp4">
   Your browser does not support video.
</video>

<!-- AUDIO -->
<h3>Audio Example</h3>
<audio controls>
   <source src="audio.mp3" type="audio/mpeg">
</audio>


<!-- DATALIST -->
<h3>Datalist Example</h3>
```

```
<input list="browsers" placeholder="Choose browser">
<datalist id="browsers">
   <option value="Chrome">
   <option value="Firefox">
   <option value="Edge">
</datalist>

<!-- BDI -->
<h3>BDI Example</h3>
<p>User <bdi class="name">الأعشى</bdi>: 1st place</p>
<br>

<!-- OUTPUT -->
<h3>Output Example</h3>
<form oninput="result.value=parseInt(a.value)+parseInt(b.value)">
   <input type="number" id="a" value="0"> +
   <input type="number" id="b" value="0"> =
   <output name="result">0</output>
</form>

<!-- FOOTER -->
<footer>  <p>Footer Section</p></footer>

</body>
</html>
```

OUTPUT:



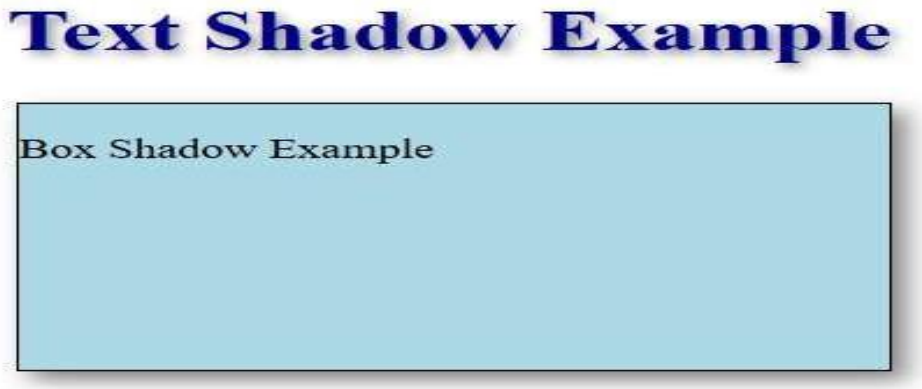
CASCADING STYLESHEET

1. Create Box Shadow and text Shadow using CSS3.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Shadow</title>
   <style>
      /* Text Shadow */
      h1 {
         color: darkblue;
         text-shadow: 2px 2px 5px gray;
      }
      /* Box Shadow */
      .box {
         width:  300px;
         height: 150px;
         margin: auto;
         background-color: lightblue;
         border: 1px solid black;
         /* Box Shadow Syntax */
         box-shadow: 5px 5px 10px gray;
      }
   </style>
</head>

<body>
<h1>Text Shadow Example</h1>
<div class="box">
   <p>Box Shadow Example</p>
</div>

</body>
</html>
```

OUTPUT:

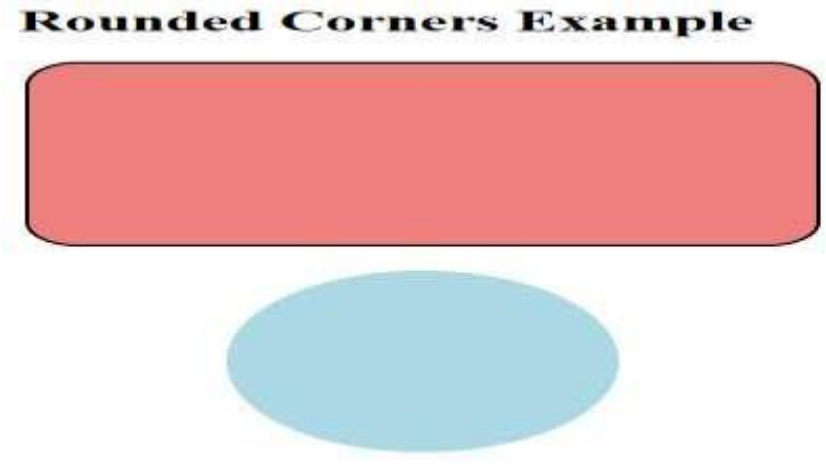


2. Create Rounded Corners using css3.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Rounded Corners</title>
   <style>
      /* Rounded Box */
      .box {
         width: 300px;
         height: 150px;
         background-color: lightcoral;
         margin: auto;
         border: 2px solid black;

         /* Rounded Corners */
         border-radius: 20px;
      }
      /* Circle Example */
      .circle {
         width: 150px;
         height: 150px;
         background-color: lightblue;
         margin: 20px auto;

         /* Full Rounded */
         border-radius: 50%;
      }
   </style>
</head>

<body>
<h2>Rounded Corners Example</h2>
<div class="box"></div>
<div class="circle"></div>

</body>
</html>
```

OUTPUT:



3. Create a web page to show fixed background image (this image will not scroll with the rest of the page)

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Fixed Background Image</title>
   <style>
      body {
         margin: 0;
         font-family: Arial;
         /* Background Image */
         background-image: url("../images/image.png");
         /* Do not repeat */
         background-repeat: no-repeat;
         /* Cover full screen */
         background-size: cover;
         /* FIXED background */
         background-attachment: fixed;
         /* Center the image */
         background-position: center;
      }
      /* Content to enable scrolling */
      .content {
         padding: 20px;
         height: 2000px;
         color: white;
      }
   </style>
</head>

<body>
<div class="content">
   <h2>Fixed Background Image Example</h2>
   <p>Scroll the page... background will not move.</p>
   <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit...</p>
   <p>More content here to enable scrolling...</p>
</div>
</body>
</html>
```

OUTPUT:

4. Create a web page to position a background image and repeat the image horizontally or vertically.

CODE:

```
<!DOCTYPE html>
<html >
<head>
   <title>Background Repeat Example</title>
   <style>
      body {
         margin: 0;
         padding: 0;
         font-family:  sans-serif;
      }
      /* Container to demonstrate horizontal repeat */
      .horizontal-repeat {
         height: 300px;
         border-bottom: 2px solid #333;
         background-image: url('../images/w3school.jpeg');
         background-repeat: repeat-x; /* Repeats horizontally only */
         background-position: top left;
      }
      /* Container to demonstrate vertical repeat */
      .vertical-repeat {
         height: 500px;
         background-image: url('../images/w3school.jpeg');
         background-repeat: repeat-y; /* Repeats vertically only */
         background-position: center top; /* Positioned in the middle-top */
      }
   </style>
</head>
<body>
   <div  class="horizontal-repeat">
      <h2 style="color: azure;">Repeat Horizontal (repeat-x)</h2>
   </div>
   <div  class="vertical-repeat">
      <h2 style="color: rgb(13, 238, 238);">Repeat Vertical (repeat-y)</h2>
   </div>
</body>
</html>
```

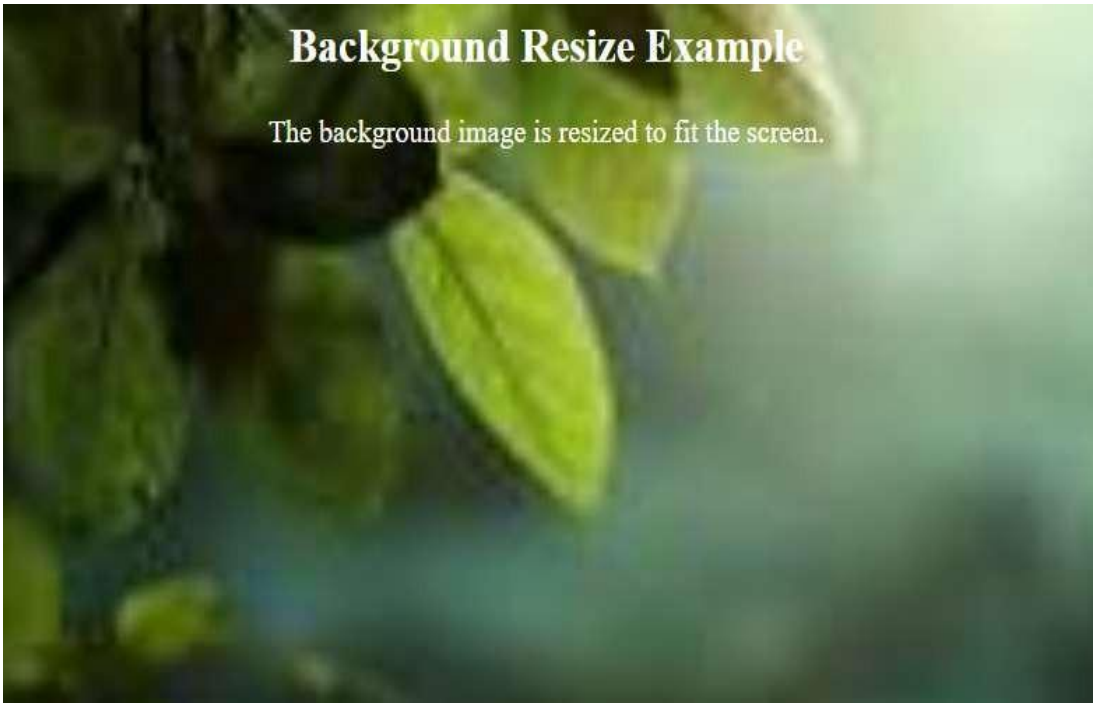
OUTPUT:

5. Create a web page to resize the background of web page.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Resize Background Image</title>
   <style>
      body {
         text-align: center;
         color: white;
         /* Background Image */
         background-image: url("../images/backgroung.jpeg");

         /* Resize background */
         background-size: cover;  /* covers full screen */
         background-attachment: fixed;
      }
      .content {
         height: 100%;
      }
   </style>
</head>

<body>
<div class="content">
   <h2>Background Resize Example</h2>
   <p>The background image is resized to fit the screen.</p>
</div>
</body>
</html>
```

OUTPUT:



6. Create a web page to show all hyperlinks with following specification: Default color is pink. Active color is blue Visited color is Green Hyperlink should be without underline.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Link Colors</title>
    <style>
        body {
            background-color: rgb(103, 123, 210);
        }
        a {
            font-size: 24px;
            font-weight: bold;
        }
        a:link {
            color: pink;
            text-decoration: none;
        }
        a:visited {
            color: green;
        }
        a:active {
            color: blue;
        }
    </style>
</head>
<body>
    <a href="https://www.google.com" target="_blank">Open Google in new tab</a><br>
    <a href="https://www.facebook.com" target="_blank">Open Facebook in new tab</a><br>
    <a href="https://www.twitter.com" target="_blank">Open Twitter in new tab</a><br><br>
</body>
</html>
```

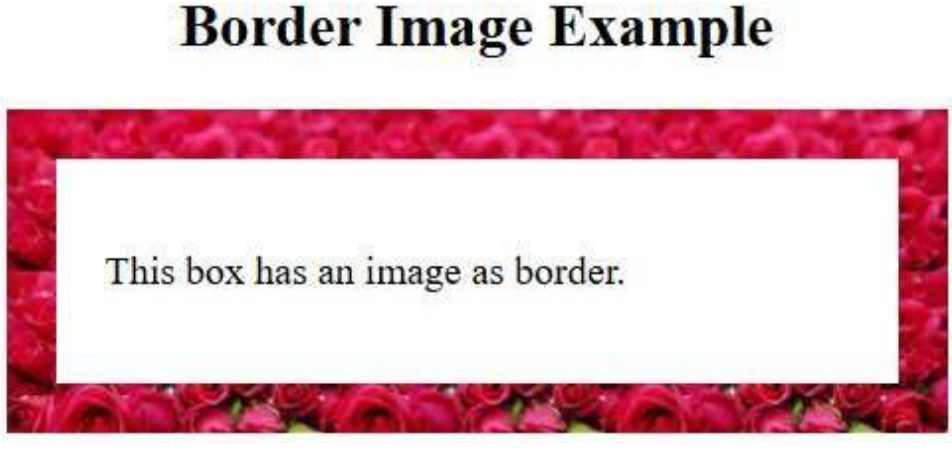
OUTPUT:

7. Design a web page to show border image.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Border Image </title>
   <style>
      .box {
         width: 300px;
         padding: 20px;
         margin: auto;

         /* Required border */
         border: 20px solid transparent;

         /* Border Image */
         border-image: url("../images/image.png") 30 round;
      }
   </style>
</head>

<body>
<h2>Border Image Example</h2>
<div class="box">
   <p>This box has an image as border.</p>
</div>
</body>
</html>
```

OUTPUT:



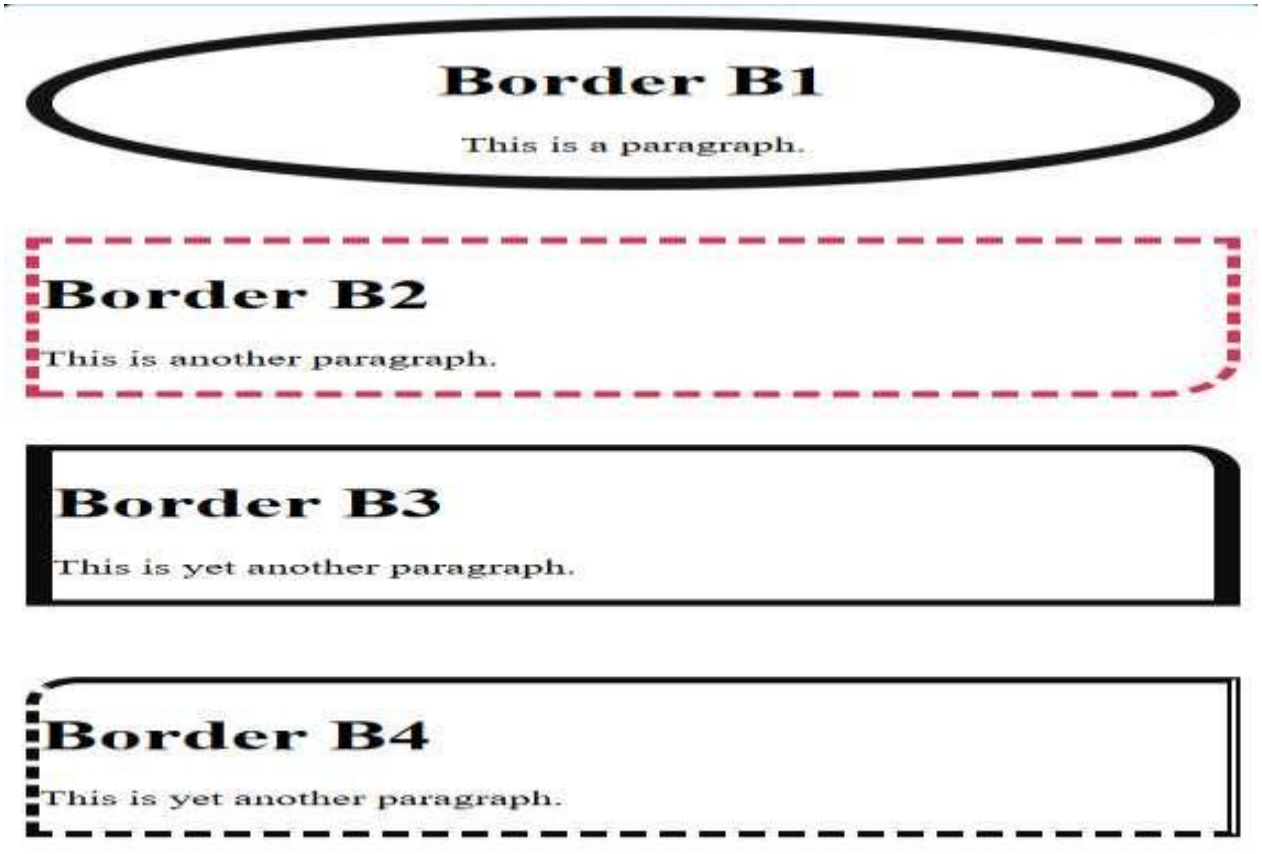
8. Design a web page to show different border style:

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
    <style>
        .a {
            border: 10px solid rgb(21, 20, 20);
            border-radius: 50%;
            text-align: center;
        }
        .b {
            border: 5px dashed rgb(220, 49, 83);
            border-bottom-right-radius: 30px;
        }
        .c {
            border-top: 5px solid rgb(12, 12, 13);
            border-right: 10px solid rgb(12, 12, 13);
            border-bottom:5px solid rgb(12, 12, 13);
            border-left:10px solid rgb(12, 12, 13);
            border-top-right-radius: 20px;
        }
        .d {
            border-right: 5px double rgb(11, 11, 11);
            border-left:5px dashed rgb(11, 11, 11);
            border-top: 5px solid rgb(11, 11, 11);
            border-bottom: 5px dashed rgb(11, 11, 11);
            border-top-left-radius: 20px;
        }
        </style>
</head>
<body>
    <div class="a">
        <h1>Border B1</h1>
        <p>This is a paragraph.</p>
    </div><br><br>
    <div class="b">
        <h1>Border B2</h1>
        <p>This is another paragraph.</p>
    </div><br><br>
    <div class="c">
        <h1>Border B3</h1>
```

```
        <p>This is yet another paragraph.</p>
    </div><br><br><br>
    <div class="d">
        <h1>Border B4</h1>
        <p>This is yet another paragraph.</p>
    </div>
</body>
</html>
```

OUTPUT:



9. **Use different font styles**

- **Set background image for both the page and single elements on page.**
- **Control the repetition of image with background-repeat property**
- **Add customized cursors for links.**
- **Work with layers.**

CODE:

```
<!DOCTYPE html>
    <html>
    <head>
      <title>Web Page </title>
      <style>
        body {
          margin: 0;
          text-align: center;
          /* Page Background */
          background-image: url("../images/w3school.jpeg");
          background-repeat: no-repeat;
          background-size: cover;
          font-family: Arial;
        }
        h2 {
          font-family: Georgia;
          color: darkgrey;
        }
        /* Banner with repeating image */
        .banner {
          background-image: url("../images/image.png");
          background-repeat: repeat-x;   /* horizontal repeat */
          color: white;
          padding: 15px;
          font-family: "Courier New";
        }
        /* Custom cursor for links */
        a {
          cursor: pointer;
          color: blue;
        }
        a:hover {
          cursor: crosshair;   /* custom cursor */
        }
        /* Layering */
        .layer1 {
          position: absolute;
          top: 200px;
          left: 100px;
          z-index: 1;
          opacity: 0.8;
        }

        .layer2 {
          position: absolute;
          top: 230px;
          left: 150px;
          z-index: 2;
          font-size: 30px;
```

```
            font-family: Verdana;
        }
        /* Element background */
        .box {
            width: 300px;
            height: 150px;
            margin: 50px auto;
            background-image: url("../images/youtube.jpeg");
            background-repeat: repeat-y;  /* vertical repeat */
            border: 2px solid black;
        }
    </style>
</head>

<body>
<h2>Web Page using CSS</h2>
<div class="banner">
    This is banner with repeating background image
</div>
<a href="#">Register</a>
<!-- Layer Example -->
<div class="layer1">
    <img src="../images/image.png" width="200">
</div>
<div class="layer2">
    <b>LAYER 2</b>
</div>
<!-- Background on element -->
<div class="box">
    <p>Background repeated vertically</p>
</div>

</body>
</html>
```

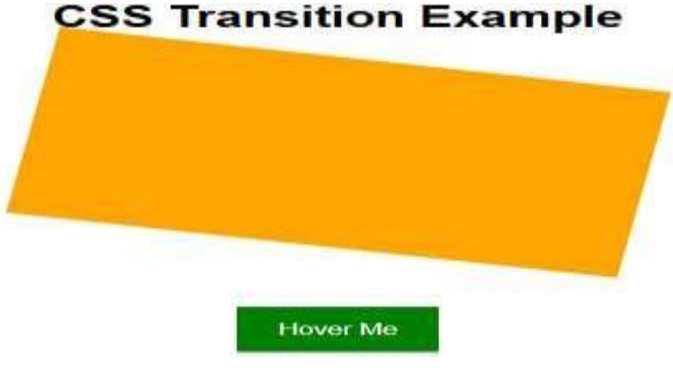
OUTPUT:

9. Create a web page to show transition effect in such a way so that elements gradually change from one style to another style.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Transition Effect</title>
   <style>
      /* Box with transition */
      .box {
         width: 200px;
         height: 100px;
         background-color: lightblue;
         margin: auto;
         /* Transition property */
         transition: all 0.5s ease;
      }
      /* Hover effect (change style) */
      .box:hover {
         width: 300px;
         height: 150px;
         background-color: orange;
         transform: rotate(10deg);
      }
      /* Button transition */
      button {
         padding: 10px 20px;
         margin-top: 30px;
         background-color: green;
         color: white;
         border: none;
         transition: background-color 0.5s;
      }
      button:hover {
         background-color: red;
         cursor: crosshair;
      }
   </style>
</head>
<body>
<h2>CSS Transition Example</h2>
<div class="box"></div>
<br>
<button>Hover Me</button>
</body>
</html>
```

OUTPUT:



JAVASCRIPT

1. **Design a web page to validate credit card number according to the below specification. Following tables outlines the major credit cards you might want to validate, along with their allowed prefixes and lengths.**

Major Credit Cards, Their Prefixes, and Length

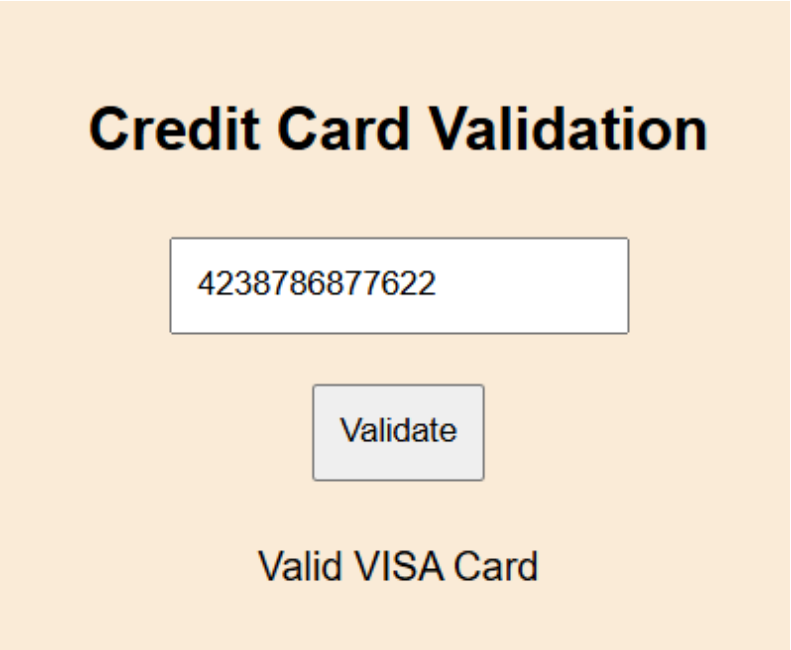
Card Type	Prefixes	Length
Master Card	51-55	16
Visa	4	13,16
American Express	34,37	15

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Credit Card Validation</title>

  <style>
    body {
      text-align: center;
      font-family: Arial;
      margin-top: 50px;
      background-color: antiquewhite;
    }
    input, button {
      padding: 10px;
      margin: 10px;
    }
  </style>
  <script>
    function validateCard() {
      let card = document.getElementById("card").value;
      let result = "";

      // VISA: starts with 4, length 13 or 16
      if ((card.startsWith("4")) CC (card.length == 13 || card.length == 16)) {
        result = "Valid VISA Card";
      }

      // MasterCard: starts 51-55, length 16
      else if ((card.length == 16) CC
          (card.startsWith("51") || card.startsWith("52") ||
           card.startsWith("53") || card.startsWith("54") ||
           card.startsWith("55"))) {
        result = "Valid MasterCard";
      }

      // American Express: starts 34 or 37, length 15
      else if ((card.length == 15) CC
          (card.startsWith("34") || card.startsWith("37"))) {
        result = "Valid American Express Card";
```

```
        }

        else {
           result = "Invalid Card Number";
        }

        document.getElementById("output").innerHTML = result;
     }
  </script>
</head>

<body>

<h2>Credit Card Validation</h2>

<input type="text" id="card" placeholder="Enter card number">

<br>

<button onclick="validateCard()">Validate</button>

<p id="output"></p>

</body>
</html>
```

OUTPUT:



2. **Write a java script to validate the following fields in a registration page:**

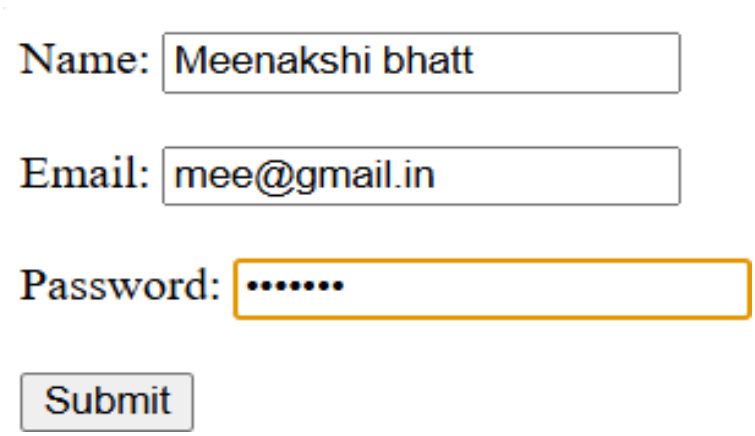
- Name (should contains alphabets and the length should not be less than 6 characters)
- Password(should not be less than 6 characters)
- E-mail(should not contain invalid addresses)

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Document</title>
</head>
<body>
  <script>
    function displayMessage() {
      let name = document.getElementById("n1").value;
      let email = document.getElementById("n2").value
      let password = document.getElementById("n3").value;
      if (name ==" "CC email=="" CC password==="") {
        alert("All fields are required.");
        return false;
      }
      let namePattern = /^[A-Za-z]+$/;

      if (!name.match(namePattern) || name.length < 6) {
        alert("Name must contain only alphabets and be at least 6 characters long");
        return false;
      }

      // Password validation
      if (password.length < 6) {
        alert("Password must be at least 6 characters long");
        return false;
      }

      // Email validation
      let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

      if  (!email.match(emailPattern)) {
        alert("Invalid Email Address");
        return false;
      }

      alert("Form Submitted Successfully!");
      return true;
    }

  </script>
  <form onsubmit="return displaymessage()">
    <label for="name">Name:</label>
    <input type="text" id="n1" name="name"  placeholder="Enter your name " required autofocus>
    <br><br>

    <label for="email">Email:</label>
    <input type="email" id="n2" name="email" placeholder="Enter your email" required>
```

```
    <br><br>

    <label for="password">Password:</label>
    <input type="password" id="n3" name="password" placeholder="Enter your password" required
maxlength="20">
    <br><br>

    <button type="submit">Submit</button>
  </form>
</body>
</html>
```

OUTPUT:

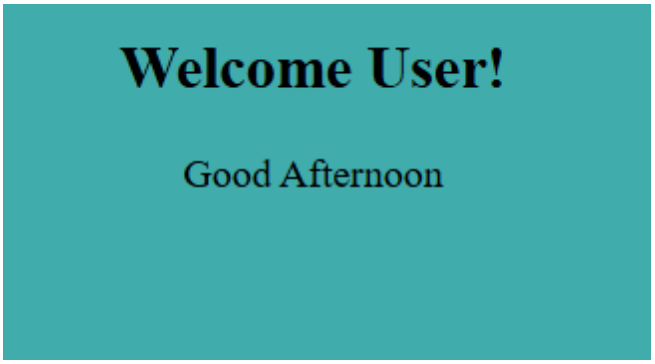


3. **Write a java script program to "Wish a user" at different hours of a day.**

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Greeting Program</title>
  <script>
    function wishUser() {
      let hour = new Date().getHours();
      let message = "";
      if (hour < 12) {
        message = "Good Morning";
      }
      else if (hour < 18) {
        message = "Good Afternoon";
      }
      else {
        message = "Good Evening";
      }
      document.getElementById("result").innerHTML = message;
    }
  </script>
</head>
<bodystyle="background-color: rgb(65, 172, 172);text-align:center;”
 onload="wishUser()" style="text-align:center; margin-top:50px;">

<h2>Welcome User!</h2>
<p id="result"></p>
</body>
</html>
```

OUTPUT:



4. Design a web page that self-modifying itself after every one minute.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Self Modifying Page</title>
</head>
<body>
<h2 id="content">Welcome to My Web Page</h2>
<script>
function changeContent() {
  let text = document.getElementById("content");
  if (text.innerHTML === "Welcome to My Web Page") {
    text.innerHTML = "Content Updated After 1 Minute!";
  } else {
    text.innerHTML = "Welcome to My Web Page";
  }
}
// Change content every 1 minute (60000 ms)
setInterval(changeContent, 6000);
</script>
</body>
</html>
```

OUTPUT:

Content Updated After 1 Minute!

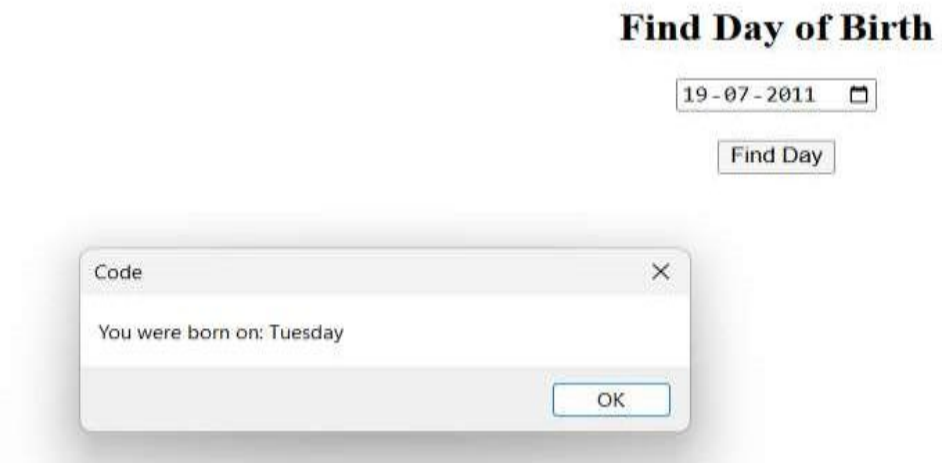
Welcome to My Web Page

5. Write an code for web application, which accepts the birth date from the user in a textbox and display the day of the week in a message box on the click of a button.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Find Day of Birth</title>
  <script>
    function findDay() {
      let dob = document.getElementById("dob").value;

      if (dob == "") {
        alert("Please enter your birth date");
        return;
      }
      // Convert input to Date object
      let date = new Date(dob);

      // Array of days
      let days = ["Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"];

      // Get day index (0-6)
      let dayIndex = date.getDay();

      // Show result
      alert("You were born on: " + days[dayIndex]);
    }
  </script>
</head>
<body style="text-align:center; margin-top:50px;">
<h2>Find Day of Birth</h2>
<input type="date" id="dob">
<br><br>
<button onclick="findDay()">Find Day</button>
</body>
</html>
```

OUTPUT:



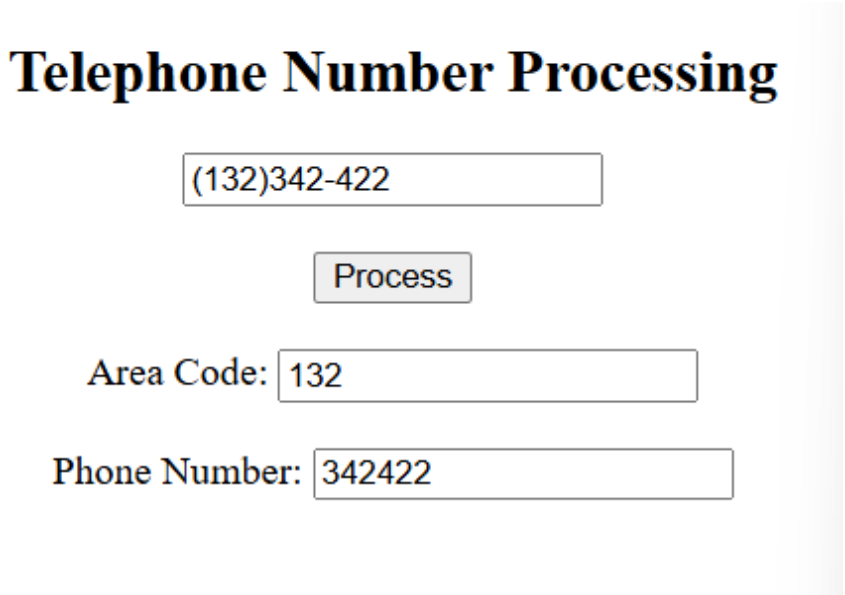
6. **Write a script that inputs a telephone number as a string in the form (555)555-555. The script should use strings method split to extract the area code as token and the last four digits of the phone numbers as a token. Display the area code in one test field and the seven digit phone number in another text field.**

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Phone Number Split</title>
  <script>
    function processNumber() {
      let phone = document.getElementById("phone").value;
      // Split by ')'
      let parts1 = phone.split(")");

      // Extract area code (remove '(')
      let areaCode = parts1[0].replace("(", "");

      // Split remaining part by '-'
      let parts2 = parts1[1].split("-");

      let firstPart = parts2[0];  // 555
      let lastPart = parts2[1];    // 555

      let fullNumber = firstPart + lastPart;  // 555555

      // Display results
      document.getElementById("area").value = areaCode;
      document.getElementById("number").value = fullNumber;
    }
  </script>
</head>

<body  style="text-align:center;  margin-top:50px;">
```

```
<h2>Telephone Number Processing</h2>

<!-- Input -->
<input type="text" id="phone" placeholder="(555)555-555">
<br><br
<button onclick="processNumber()">Process</button>
<br><br>

<!-- Output -->
<label>Area Code:</label>
<input type="text" id="area" readonly>
<br><br>
<label>Phone Number:</label>
<input type="text" id="number" readonly>

</body>
</html>
```

OUTPUT:



7. **Create a web page that applies the invert filter to an image if the user moves the mouse over it.**

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Image Hover Effect</title>
  <style>
    img {
      width: 300px;
      transition: 0.5s;
    }

    img:hover {
      filter: invert(100%);
    }
  </style>
</head>
<body>

<h2>Hover on Image</h2>

<img src="https://img.freepik.com/free-photo/closeup-scarlet-macaw-from-side-view-scarlet-macaw-closeup-head_488145-3540.jpg?semt=ais_hybridCw=740Cq=80" alt="Sample Image">

</body>
</html>
```

OUTPUT:

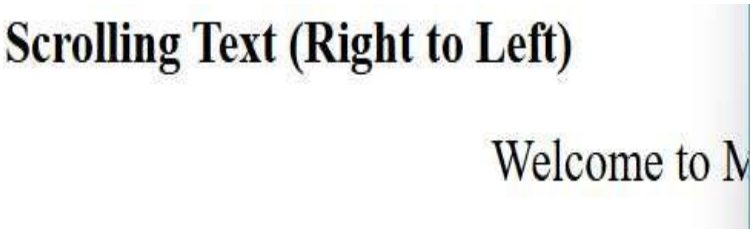
Hover on Image

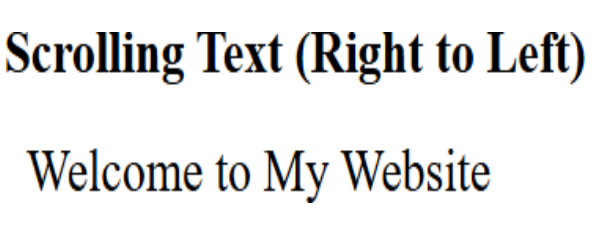
8. **Write an HTML Page to scroll the text message from right to left at the status bar of browser window.**

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Scrolling Text</title>
  <style>
    #scrollText {
      position: relative;
      font-size: 24px;
    }
  </style>
  <script>
    let position = window.innerWidth;
    function scrollText() {
      let text = document.getElementById("scrollText");

      position--;

      text.style.left = position + "px";

      if (position < -text.offsetWidth) {
        position = window.innerWidth;
      }

      setTimeout(scrollText, 2);
    }
  </script>
</head>
<body onload="scrollText()">
<h2>Scrolling Text (Right to Left)</h2>
<div id="scrollText">Welcome to My Website</div>
</body>
</html>
```

OUTPUT:





G. Design a web page to perform survey on four different model of Maruti (Maruti - K10, Zen-Astelo, Wagnor, Maruti- SX4) owned by person living in four metro cities(Delhi, Mumbai, Chennai sKolkatta). Display tabulated report like format given below:

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Maruti Car Survey</title>

  <script>
    function calculate() {
      let cities = ["Delhi", "Mumbai", "Chennai", "Kolkata"];
      let models = ["k10", "zen", "wagon", "sx4"];

      for (let i = 0; i < cities.length; i++) {
        let rowTotal = 0;

        for (let j = 0; j < models.length; j++) {
          let id = cities[i] + "_" + models[j];
          let value = parseInt(document.getElementById(id).value) || 0;
          rowTotal += value;
        }

        document.getElementById(cities[i] + "_total").innerHTML = rowTotal;
      }
    }
  </script>
</head>

<body>

<h2 align="center">Maruti Car Survey</h2>

<table border="1" align="center" cellpadding="10">
  <tr>
    <th>City</th>
    <th>Maruti-K10</th>
    <th>Zen-Astelo</th>
    <th>Wagnor</th>
    <th>Maruti-SX4</th>
    <th>Total</th>
  </tr>

  <!-- Delhi -->
  <tr>
    <td>Delhi</td>
    <td><input type="number" id="Delhi_k10"></td>
    <td><input type="number" id="Delhi_zen"></td>
    <td><input type="number" id="Delhi_wagon"></td>
    <td><input type="number" id="Delhi_sx4"></td>
    <td id="Delhi_total"></td>
  </tr>

  <!-- Mumbai -->
  <tr>
```

```
    <td>Mumbai</td>
    <td><input type="number" id="Mumbai_k10"></td>
    <td><input type="number" id="Mumbai_zen"></td>
    <td><input type="number" id="Mumbai_wagon"></td>
    <td><input type="number" id="Mumbai_sx4"></td>
    <td id="Mumbai_total"></td>
  </tr>

  <!-- Chennai -->
  <tr>
    <td>Chennai</td>
    <td><input type="number" id="Chennai_k10"></td>
    <td><input type="number" id="Chennai_zen"></td>
    <td><input type="number" id="Chennai_wagon"></td>
    <td><input type="number" id="Chennai_sx4"></td>
    <td id="Chennai_total"></td>
  </tr>

  <!-- Kolkata -->
  <tr>
    <td>Kolkata</td>
    <td><input type="number" id="Kolkata_k10"></td>
    <td><input type="number" id="Kolkata_zen"></td>
    <td><input type="number" id="Kolkata_wagon"></td>
    <td><input type="number" id="Kolkata_sx4"></td>
    <td id="Kolkata_total"></td>
  </tr>
</table>

<br>

<center>
  <button onclick="calculate()">Calculate Totals</button>
</center>

</body>
</html>
```

OUTPUT:

Maruti Car Survey

City	Maruti-K10	Zen-Astelo	Wagnor	Maruti-SX4	Total
Delhi	4	12	31	12	59
Mumbai	24	14	22	53	113
Chennai	34	42	14	24	114
Kolkata	12	23	14	42	91

Calculate Totals

PHP

1. Create a Login form having User-Id and Password fields. After submitting the form match the user-id and password with existing user-id and password. If user-id and password match a new welcome window should be appear.

```
<?php
$valid_user = "admin";
$valid_pass = "12345";

$isLoggedIn = false;
$message = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $user = $_POST["userid"];
    $pass = $_POST["password"];

    if ($user == $valid_user && $pass == $valid_pass) {
        $isLoggedIn = true;
    } else {
        $message = "Invalid User-Id or Password";
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Login Form</title>
</head>
<body>

<?php if ($isLoggedIn) { ?>
    <h2>Welcome <?php echo htmlspecialchars($user); ?>!</h2>
<?php } else { ?>

    <h2>Login Form</h2>

    <form method="post">
        User-Id:
        <input type="text" name="userid" required><br><br>

        Password:
        <input type="password" name="password" required><br><br>

        <input type="submit" value="Login">
    </form>

    <p style="color:red;"><?php echo $message; ?></p>

<?php } ?>

</body>
</html>
```